Matrix Operations at a Bike-Share Depot

Adding, subtracting and scaling matrices, multiplying them, and writing a system as a matrix equation

- Every matrix on this sheet is a table of real quantities. The row and column meanings are stated with the matrix; the units are stated in the question.
- Adding, subtracting or scaling works entry by entry. Multiplying does not: the entry in row i, column j of a product pairs row i of the left matrix with column j of the right one.
- Write your result in the empty bracket provided, one number per position.

1. A bike-share program runs two depots. Rentals are counted in bikes, with rows North depot then South depot, and columns standard then electric.

$$\text{Monday } M = \begin{bmatrix} 24 & 18 \\ 31 & 12 \end{bmatrix}, \qquad \text{Tuesday } T = \begin{bmatrix} 19 & 22 \\ 27 & 15 \end{bmatrix}$$

Find $M + T$, the two-day rental total in bikes for each depot and bike type.

$$M + T = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{bmatrix}$$

2. The same program lists rental prices in dollars per rental, with rows weekday then weekend and columns standard then electric.

$$P = \begin{bmatrix} 6 & 10 \\ 5 & 9 \end{bmatrix}$$

Every price is raised by 25%, so each entry is multiplied by the scalar 1.25. Find the new price matrix $1.25P$, in dollars per rental.

$$1.25P = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{bmatrix}$$

3. Over one week, matrix U counts bikes checked out and matrix V counts bikes returned, in bikes, with rows North depot then South depot and columns standard then electric.

$$U = \begin{bmatrix} 52 & 34 \\ 47 & 29 \end{bmatrix}, \qquad V = \begin{bmatrix} 45 & 30 \\ 41 & 26 \end{bmatrix}$$

Find $U - V$, the number of bikes still out at the end of the week.

$$U - V = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{bmatrix}$$

4. Wednesday's rentals, in bikes, are $W = \begin{bmatrix} 30 & 14 \\ 22 & 20 \end{bmatrix}$ (rows North then South depot, columns standard then electric). The price column, in dollars per rental, is $c = \begin{bmatrix} 6 \\ 10 \end{bmatrix}$. Find Wc, the Wednesday revenue in dollars at each depot.

$Wc = \begin{bmatrix} \_\_\_\_ \\ \_\_\_\_ \end{bmatrix}$

5. Matrix A gives bikes added per month and matrix B gives bikes retired per month, in bikes, with rows North depot then South depot and columns standard then electric.

$$A = \begin{bmatrix} 4 & 7 \\ 2 & 5 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 3 \\ 6 & 2 \end{bmatrix}$$

The fleet grows for 3 months and then retires bikes for 2 months. Find $3A - 2B$, the net change in bikes.

$3A - 2B = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{bmatrix}$

6. Matrix D gives bikes serviced per day (rows mechanic 1 then mechanic 2, columns standard then electric). Matrix E gives parts used per bike (rows standard then electric, columns brake pads then cables).

$$D = \begin{bmatrix} 5 & 2 \\ 3 & 4 \end{bmatrix}, \qquad E = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

Find the product DE, the parts each mechanic uses per day, counted in parts.

$$DE = \begin{bmatrix} \_\_\_\_ & \_\_\_\_ \\ \_\_\_\_ & \_\_\_\_ \end{bmatrix}$$

7. Two riders buy day passes at the same depot. Two standard passes and three electric passes cost 22 dollars; four standard passes and one electric pass cost 24 dollars. Let x be the price of a standard pass and y the price of an electric pass, both in dollars.

Write the pair of conditions as a matrix equation, then find x and y.

Answer: ________________

8. Matrix F counts weekend rentals in bikes (rows Saturday then Sunday, columns standard then electric). Matrix G lists, per rental, the revenue in dollars and the staff time in minutes (rows standard then electric, columns revenue then staff minutes).

$$F = \begin{bmatrix} 8 & 3 \\ 6 & 5 \end{bmatrix}, \qquad G = \begin{bmatrix} 7 & 4 \\ 12 & 9 \end{bmatrix}$$

(a) Compute the entry of FG in row 1, column 2.

Answer: ________________

(b) Explain what that entry measures for this depot, naming what its row and its column stand for.

Answer: ________________

9. Matrix H counts one day's rentals in bikes (rows North depot then South depot, columns standard, electric, cargo). Column k lists the price per rental in dollars for the three bike types.

$$H = \begin{bmatrix} 10 & 6 & 4 \\ 8 & 9 & 3 \end{bmatrix}, \qquad k = \begin{bmatrix} 6 \\ 10 \\ 14 \end{bmatrix}$$

(a) Find Hk, the day's revenue in dollars at each depot.

Answer: ________________

(b) The product kH does not exist. Explain what about the two shapes makes it impossible.

Answer: ________________

10. A depot holds 100 bikes in three classes: standard, electric and cargo. It has 20 more electric bikes than standard bikes. If every bike in the depot were rented exactly once, at 6 dollars for a standard, 10 dollars for an electric and 14 dollars for a cargo bike, the depot would take 960 dollars. Let x, y and z be the number of standard, electric and cargo bikes.

(a) Write the three conditions as a matrix equation and find x, y and z, in bikes.

Answer: ____________________

(b) Explain what the coefficient matrix, the variable column and the constant column each stand for in this situation.

Answer: ____________________